

GenomicsOps

Variant interpretation you can audit. Offline.

Overview

GenomicsOps is a Windows desktop workbench for variant interpretation in research genomics. It runs entirely on a single workstation, uses local copies of ClinVar and gnomAD, and produces ACMG/AMP 2015 classifications that show every criterion that fired.

It is designed for labs that process PHI, work air-gapped, or simply prefer not to send variant data to a third-party service.

Research Use Only. Not for clinical diagnostic use.

Why GenomicsOps

Offline-first. All sample data, variant records, and reports live in a single SQLite database on the workstation. ClinVar and gnomAD lookups work with no network connection. The bundled NCBI MANE reference (19,363 genes) is loaded on first run.

Transparent ACMG. Every classification shows which criteria fired, what evidence supported each one, and what would change the call. There is no ML in the decision path. The engine is a deterministic function of the variant evidence and the curated criteria provided.

Gene-specific rules. 65 genes have VCEP (Variant Curation Expert

Panel) rule sets bundled, including BRCA1/2, TP53, MLH1, MSH2, MSH6, PMS2, CDH1, PTEN, APC, MUTYH, LDLR, APOB, HNF1A, HNF4A, GCK, MYH7, MYBPC3, TNNT2, KCNQ1, FBN1, GAA, PAH, and more. The engine dispatches to the gene-specific points system when one exists.

Validated. On a 12,644-variant expert-curated corpus from ClinGen ERepo, the combining engine agrees with the curator's final classification 95.8% of the time. See the Validation section for full numbers.

Tamper-evident audit log. Every action is written to a hash-chained log with SHA-256 signatures, user attribution, and timestamps. The chain can be verified from Settings → Security.

Capabilities

Variant interpretation

- VCF import (single-sample and multi-sample)
- ACMG/AMP 2015 classification engine with visible criteria
- MANE Select transcript mapping
- Local ClinVar significance lookup
- Local gnomAD population frequency lookup
- Optional Ensembl VEP consequence prediction
- HPO phenotype matching (19,500+ ontology terms)

CNV and structural variants

- Detects , <DUP>, <INV> symbolic alleles from VCF
- Scores Sections 1-3 of the ClinGen / Riggs 2020 framework automatically
- Manual entry for Sections 4-5

Family analysis

- Proband + parents trio modeling
- Inheritance pattern per variant: de novo, maternal, paternal,

- homozygous, compound heterozygous
- Feeds ACMG criteria PS2, PM6, and PP1

Cohort analysis

- Multi-sample PCA clustering
- Gene enrichment
- Shared variant analysis

Pipelines

- Bundled nf-core catalog (sarek, rnaseq, demo)
- Runs in Docker, no system Nextflow required
- Output VCFs import back as samples

Reporting and export

- PDF with per-variant ACMG evidence
- FHIR R4 DiagnosticReport + Observations
- HL7 v2 ORU^R01
- Custom JSON for LIMS

Validation

Metric	Result	Corpus
ACMG combining — overall	95.8%	12,644 curated ERepo variants
ACMG combining — pathogenic	97.9%	3,480 variants
ACMG combining — benign	98.8%	1,310 variants
Criterion firing	92.5%	80 structured-summary variants
CNV scoring (Riggs 2020 Sec. 1-3)	100%	23 curated CNV events

Trio inheritance	100%	43-variant synthetic corpus
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The design principle: the curator provides ACMG criteria; the engine combines them. Automated extraction of criteria from narrative summaries is not attempted — that path has a 2.8% F1 in published work, and is not how the workflow is designed to be used.

Deployment and data handling

All data stays on the workstation. The application has no cloud component. It does not phone home, does not collect telemetry, and does not require a license server to run.

Only optional enrichment (Ensembl VEP for consequence prediction and PubMed for literature) uses network. Both can be disabled in Settings. The bundled local ClinVar and gnomAD databases provide the core annotation offline.

System requirements

Component	Requirement
OS	Windows 10 (19045+) or Windows 11
Architecture	x64
RAM	8 GB recommended
Installer size	~220 MB
Reference data	~6.5 GB (ClinVar + gnomAD)
WebView2	Included with supported Windows builds

Licensing

Licenses are cryptographically signed tokens bound to the machine fingerprint. Activation is offline: the customer copies the fingerprint from the License page, sends it to us, and pastes back the returned token. No phone-home, no license server, no per-launch checks.

One license per machine. Volume pricing is available for 5 or more seats. Contact sales@genomicsops.io.

Who it's for

- **Academic genomics research labs** — run the interpretation pipeline on a workstation without depending on external services.
- **Bioinformatics core facilities** — give PIs a self-service workbench with a shared ACMG engine and audit trail.
- **Biotech R&D teams** — keep candidate variant data on-prem with reproducible, exportable classifications.
- **Clinical labs (research use)** — evaluate the ACMG engine against your own curated corpus before adopting for research pipelines.

Who it's not for

- Clinical diagnostic workflows
 - Full genome alignment or variant calling
 - FDA-cleared or CE-IVD software requirements
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Contact

- Sales: sales@genomicsops.io
- Support: support@genomicsops.io
- Security: security@genomicsops.io

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